

Power-BI

Project-1-Student Marks Dashboard step-by-step

STEP 1: Open Microsoft Excel

- ❖ Open **Microsoft Excel** on your computer.
- ❖ Create a **new blank workbook**.

STEP 2: Enter the Data

In your Excel sheet, type exactly the following in cells A1 to C7:

A	B	C
Name	Subject	Marks
Alice	Math	85
Alice	English	78
Bob	Math	92
Bob	English	81
Charlie	Math	68
Charlie	English	74

STEP 3: Save the Excel File

Follow these steps:

1. Click on **File** in the top-left corner of Excel.
2. Click on **Save As** (or **Browse** if needed).
3. Choose a folder (like Desktop or Documents).
4. In the **File name** box, type:
StudentMarks
5. Make sure the **Save as type** is:
Excel Workbook (*.xlsx)
6. Click **Save**.

The file StudentMarks.xlsx is now ready for Power BI.

STEP 4: Open Power BI and Load the Data

Step-by-step instructions:

1. **Open Power BI Desktop**
(If you don't have it, you can download it for free from <https://powerbi.microsoft.com>)
2. In Power BI, click **Home > Get Data > Excel**
(Top-left corner)
3. A file browser will open.
 - ❖ Navigate to where you saved StudentMarks.xlsx
 - ❖ Click the file and then click **Open**
4. A Navigator window will appear:
 - ❖ You'll see a preview of your data.
 - ❖ Make sure the checkbox next to your table/sheet is ticked.
 - ❖ Click **Load**

Power BI will now load the data into your workspace.

STEP 5: Create a Bar Chart for Average Marks per Student

Follow these steps:

- ❖ In Power BI, go to the **Report view**
(You'll see a white canvas with a **Visualizations** panel on the right)
- ❖ In the **Visualizations** panel, click on the **Clustered Column Chart**
(It's the icon that looks like 3 vertical bars)

- ❖ A blank chart box will appear on the canvas.
- ❖ On the **right panel**, under **Fields**:
 - ❖ Drag Name and drop it into the **X-axis** area.
 - ❖ Drag Marks and drop it into the **Y-axis / Values** area.
- ❖ By default, it shows **Sum of Marks**.
 - ✓ Click the small arrow next to Sum of Marks in the **Values** field.
 - ✓ Choose **Average** from the dropdown.
- ❖ Click on the **title** of the chart (double-click it) and rename it to:
"Average Marks per Student"

 You now have your first chart!

STEP 6: Create Subject-wise Marks Chart by Student

Follow these steps:

1. Click on a **blank area** of the canvas to deselect the first chart.
2. In the **Visualizations** panel, click on **Stacked Bar Chart**
 (It looks like 3 horizontal bars)
3. A new blank chart box will appear.
4. In the **Fields** panel on the right:
 - ❖ Drag Subject into the **Y-axis** (or Axis)
 - ❖ Drag Marks into the **X-axis** (or Values)
 - ❖ Drag Name into the **Legend**
5. Double-click the chart title and rename it to:
"Marks by Subject and Student"

You should now see colored bars comparing each student's marks in Math and English.

STEP 7: Create KPI Cards for Highest and Lowest Marks

A. Create the Highest Mark Card

1. Click on a blank area of the canvas.
2. In the **Visualizations** panel, click the **Card** visual
(Icon looks like a big number in a rectangle □)
3. A blank card appears.
4. In the **Fields** panel:
 - ❖ Drag Marks into the **Values** section of the card.
5. By default, it shows **Sum of Marks**.
 - ❖ Click the dropdown next to Sum of Marks
 - ❖ Choose **Maximum**
6. Double-click the title of the card and rename it to:
"Highest Mark"

B. Create the Lowest Mark Card

1. Click on a new blank space on the canvas.
2. Again, select the **Card** visual from the panel.
3. Drag Marks into the **Values** field.
4. Change the aggregation:
 - ❖ Click the dropdown next to Sum of Marks
 - ❖ Choose **Minimum**
5. Rename the title to:
"Lowest Mark"

Now you should have two cards showing the highest (e.g., 92) and lowest (e.g., 68) marks.

STEP 8: Add Slicers (Filters)

A. Add a Slicer for Name (Student)

1. Click on a **blank area** of the canvas.
2. In the **Visualizations** panel, click the **Slicer** icon
(It looks like a funnel with a list)
3. In the **Fields** panel, drag Name into the slicer.
4. You'll now see a list of student names (Alice, Bob, Charlie) as a filter.

You can now **click on a name**, and all visuals will update to show only that student's data.

B. (Optional) Add a Slicer for Subject

1. Click on another blank space on the canvas.
2. Insert another **Slicer** (same way as above).
3. Drag Subject into it.

You can now filter the dashboard by subject (Math or English).

STEP 9: Format and Style Your Dashboard

Follow these steps to make your visuals more readable and appealing:

A. Rename All Visual Titles

1. Click on each chart or card one by one.
2. In the **Visualizations panel**, go to the **Format** section (paint roller icon  .
3. Expand the **Title** section:
 - ❖ Toggle “Title” ON (if not already)
 - ❖ Edit the **Title Text** to match the chart:
 - “Average Marks per Student”
 - “Marks by Subject and Student”
 - “Highest Mark”
 - “Lowest Mark”

B. Enable Data Labels for Charts

1. Click on each **bar chart** (one at a time)
2. Go to **Format (paint roller) > Data Labels**
3. Turn **Data Labels** ON

This shows the actual mark values on each bar.

C. Apply a Theme (Optional)

1. Go to the top menu: **View > Themes**
2. Select a theme like "**Colorblind Safe**" or "**Classic**"
✦ It changes all visual colors consistently.

D. Align and Resize Visuals

1. Drag each visual into a neat grid-like layout.
2. Resize visuals so nothing overlaps.
3. Arrange slicers on the side or top.

Now your dashboard looks polished and easy to read.

STEP 10: Save and Export Your Dashboard

A. Save the Power BI Project

1. Click on **File** (top-left corner)
2. Select **Save As**
3. Choose a folder (e.g., Desktop or Documents)
4. Enter the name:
StudentMarksDashboard.pbix
5. Click **Save**

This saves your entire dashboard project so you can open and edit it later in Power BI.

B. Export as PDF (Optional)

1. Go to **File > Export > Export to PDF**
2. Power BI will generate a PDF of your dashboard.
3. Save it and share with others if needed.

C. (Optional) Publish Online

If you want to share it online:

1. Click **Home > Publish**
2. Sign in with your Microsoft account
3. Choose a workspace
4. Share the online dashboard with others